6.8 Integration Using Trigonometric Substitution

Name _____ Period ____ Date _____

AP Calculus 6.8 (30-minute plan) · Student sheet

I can…

- choose the right trigonometric substitution.
- integrate and use a right triangle to return to x.
- evaluate a definite integral with a trigonometric substitution.

Vocabulary · fill in the missing word

Trigonometric substitution — _____ x with a sin θ, a tan θ, or a sec θ to remove a square root

Reference triangle — a right triangle labeled from the _____, used to write the answer in terms of x

Part 1 · Notes and examples (fill in each note, then work the example)

$\sqrt{(a^2 - x^2)}$: let x = a _____ θ.

Example 1

Find $\int dx/\sqrt{9 - x^2}$.

$\sqrt{(x^2 + a^2)}$ or $x^2 + a^2$: let x = a _____ θ.

Example 2

Find $\int dx/(x^2 + 4)$.

$\sqrt{(x^2 - a^2)}$: let x = a _____ θ.

Back to x: use a right _____ labeled from the substitution (or change the limits).

Example 3

Which substitution should be used for an integral containing $\sqrt{x^2 - 16}$?

Part 2 · Partner practice

Partner practice

(AP® Practice) $\int_0^3 \sqrt{9 - x^2}\ dx =$

(A) $\frac{9\pi}{4}$ (B) $\frac{9\pi}{2}$ (C) 3π (D) 9